

## Q1:2018 SPEC P2

8. (a) Define

(i) *Density* \_\_\_\_\_(ii) *Pressure* \_\_\_\_\_

[2]

(b) Use the definitions you have given in (a) to derive the equation  $P = \rho gh$  for the pressure,  $P$ , at a depth,  $h$ , in a fluid of density  $\rho$ .

[1]

## Q2:NOV 2008 P5

3 (a) (i) Define the terms *density* and *pressure*.(ii) Derive the equation  $P = \rho gh$ , where the symbols have their usual meaning.

[4]

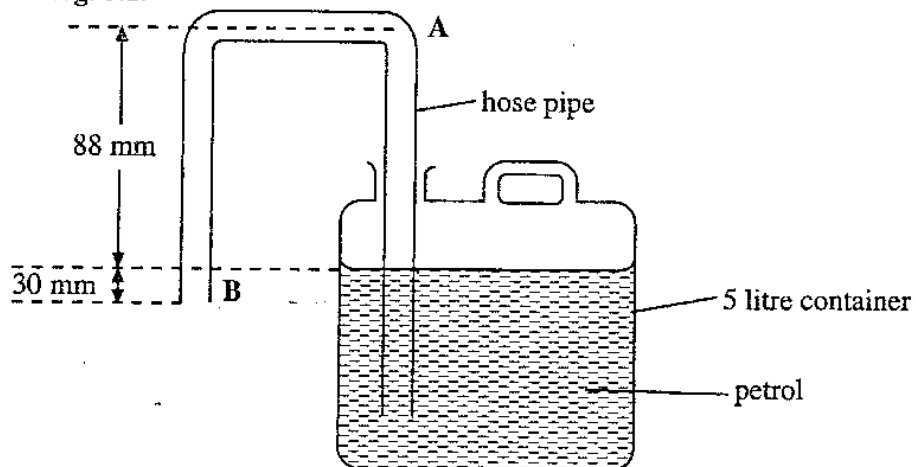
(b) At atmospheric pressure of  $1.01 \times 10^5$  Pa, a hose pipe is arranged to drain petrol of density  $0.900 \text{ g/cm}^3$  from a five litre container as shown in Fig. 5.1.

Fig. 5.1

The hose pipe is first filled with petrol and the end B is closed with a finger.

6.1 PHASES OF MATTER P2, P3&P5 QUESTIONS

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- (i) Describe and explain what happens if the finger is removed.
- (ii) Determine the pressure at A
  - 1 before the finger is removed,
  - 2 after the petrol stops flowing.
- (iii) Explain why the surface of the petrol is always plane and horizontal when at rest despite tilting of the container.

[8]

Q3:NOV 2011 P5

- 1 (a) (i) By using definitions of pressure and density derive the equation,  $p = \rho gh$ , with terms having their usual meanings.
- (ii) Fig. 1.1 (a) shows water in a measuring cylinder. An iron bar of mass, 1.0 kg is then held in the water as shown in Fig. 1.1 (b).

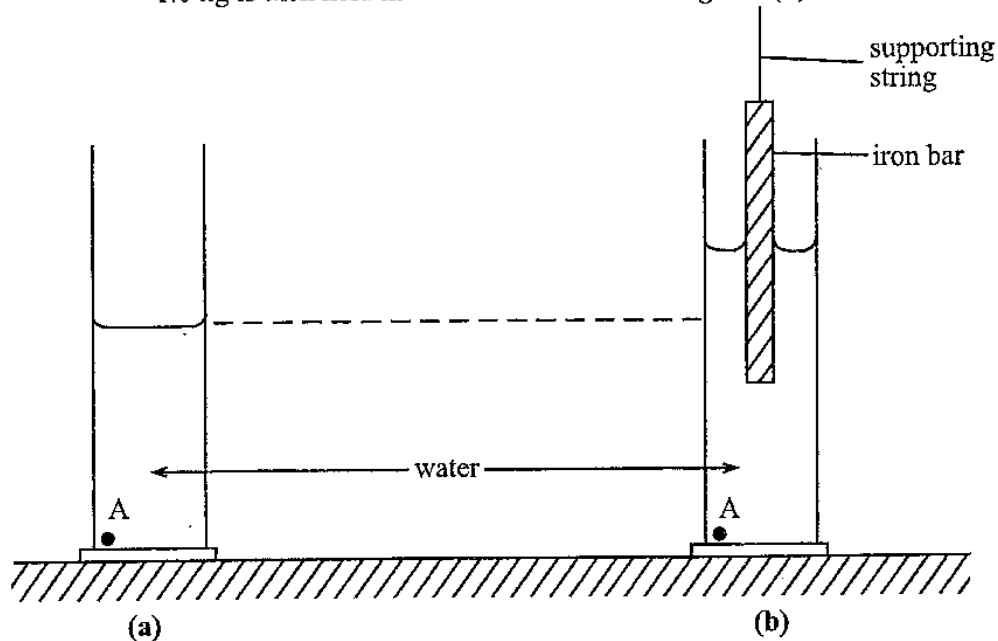


Fig. 1.1

State and explain whether the pressure at point A is different in the two situations.

[6]

## Q4: NOV 2015 P5

- 3 (a) (i) Define *density*. [3]
- (ii) Explain why the densities of ice and water are almost equal but completely different to that of steam. [3]
- (b) Outline an experiment that demonstrates Brownian motion and state the evidence provided by this experiment. [3]
- (c) Compare the movement and energies of molecules in solids, liquids and gases. [3]
- (d) On a certain day, a mercury barometer read 745 mmHg at the bottom of a mountain and 590 mmHg at the top.
- (i) Show that the height in metres, of the mountain is approximately equal to 
$$\frac{0.158 \times \text{density of mercury}}{\text{density of air}}.$$
- (ii) Give a reason why the height of the mountain is not exactly equal to the one in (i). [3]